



History of the discovery, development, and FDA-approval of buprenorphine medications for the treatment of opioid use disorder

Christian Heidbreder^{a,*}, Paul J. Fudala^a, Mark K. Greenwald^b

^a Indivior Plc, North Chesterfield, VA, United States of America

^b Department of Psychiatry and Behavioral Neurosciences, Wayne State University School of Medicine, Detroit, Michigan, United States of America

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ABSTRACT

Buprenorphine-based medications were first approved by the United States Food and Drug Administration in 2002 for the treatment of opioid dependence, or opioid use disorder (OUD) as the condition is presently known. This regulatory milestone was the outcome of 36 years of research and development, which also led to the development and approval of several other new buprenorphine-based medications. In this short review, we first describe the discovery and early development stages of buprenorphine. Second, we review key steps that led to the development of buprenorphine as a drug product. Third, we explain the regulatory approval of several buprenorphine-based medications for the treatment of OUD. We also discuss these developments in the context of the evolution of regulations and policies that have progressively improved OUD treatment availability and efficacy, although challenges remain in removing system-level, provider-level, and local-level barriers to quality treatment, to integrating OUD treatment into routine care and other settings, to reducing disparities in access to treatment, and to optimizing person-centered outcomes.

1. Introduction

In 2020, an estimated 9.5 million Americans aged 12 or older mis-used opioids, 2.7 million met criteria for opioid use disorder (OUD) (American Psychiatric Association, 2013), and 278,000 received treatment in the past year for OUD (SAMHSA, 2021a). The opioid crisis has also been compounded by the coronavirus disease 2019 (COVID-19) pandemic as well as co-use or contamination of psychostimulants with illicitly manufactured fentanyl or heroin (Jones et al., 2020; Niles et al., 2021). Individuals with OUD are at higher risk for COVID-19 (Wang et al., 2021) among patients with a substance use disorder and are particularly vulnerable to treatment interruption, isolation, and stress, all of which can trigger misuse of prescribed opioids, illicit opioid use, and return to opioid use (Rodda et al., 2020). Early in the pandemic, drug testing revealed increased positivity rates for non-prescribed fentanyl (+35%), heroin (+44%), and opioids (+10%) (Jones et al., 2020). Since then, multiple studies in the United States (US) have shown increases in opioid-related mortality and emergency department visits (Wang et al., 2021; Holland et al., 2021; Slavova et al., 2020). The predicted provisional number of opioid overdose deaths in the US increased to 81,857 in the 12-month period ending in February 2022, up from 72,930 the year before in February 2021 (CDC, 2022). Synthetic opioids alone accounted for 72,758 overdose deaths in the US during the same period (CDC, 2022). More than ever, it is critical

that individuals suffering from OUD receive efficacious, evidence-based treatments.

A significant breakthrough in the treatment of opioid dependence, or OUD as the disorder is presently known (American Psychiatric Association, 2013), occurred in the US during the 1960s with the expansion of methadone treatment and a publication by Dole and Nyswander describing the treatment of chronic heroin users with orally-administered methadone (Dole and Nyswander, 1965; Jaffe and O'Keefe, 2003). However, the Federal Register methadone regulations of 1972 (US FDA, 1972) and the Narcotic Addict Treatment Act (NATA, 1974) of 1974 limited methadone maintenance treatment to opioid-treatment-programs (OTPs) – “methadone-clinic” settings (Institute of Medicine, 1995) – and thus established a closed and highly-regulated distribution system for methadone such that patients could only receive methadone treatment in the context of an OTP. This restricted OTP system was geographically biased toward urban centers, with greater representation of economically disadvantaged individuals, Blacks, and other minorities, which regrettably contributed to racial disparities and stigma in treatment for OUD (Andraka-Christou, 2021; Gryczynski et al., 2011; Hansen et al., 2016; Pro-and Zaller 2020; Stein et al., 2018). A transformational and radical move to bring OUD treatment to the mainstream of medical care took place in October 2000 with the passage of the Children's Health Act (P.L. 106–310), with Title XXXV of the Act providing a “Waiver Authority for Physicians Who Dispense or Prescribe Cer-

* Corresponding author.

E-mail address: Christian.Heidbreder@indivior.com (C. Heidbreder).

tain Narcotic Drugs for Maintenance Treatment or Detoxification Treatment of Opioid-Dependent Patients.” This part of the law is commonly known as the Drug Addiction Treatment Act of 2000 (DATA 2000) (DATA 2000, 2000). Under the provisions of DATA 2000, qualifying physicians can obtain a waiver from the special registration requirements described in NATA (NATA, 1974), can treat OUD with Schedule III, IV, and V opioid medications that have been specifically approved by the US Food and Drug Administration (FDA) for that indication, and can prescribe/dispense these medications in treatment settings other than licensed OTPs, including office-based practices.

On July 22, 2016, President Obama signed S. 524, the Comprehensive Addiction and Recovery Act of 2016 (CARA) into law. CARA was enacted to address prevention, treatment, recovery support, criminal justice reform, overdose reversal, and law enforcement (CARA, 2016). CARA and the Office of National Drug Control Policy (ONDCP) Reauthorization Act (ONDCP Reauthorization Act of 2006, 2006) also amended Section 303(g)(2) of the Controlled Substances Act (21 USC 823(g)(2)) to increase, under certain conditions and restrictions, the total number of patients a prescriber can treat with buprenorphine from 30 to 100. Most recently, President Biden signed a multipart appropriations bill on December 29, 2022 that included the Mainstreaming Addiction Treatment (MAT) Act (H.R. 1384/S. 445) which eliminates the federal “x-waiver” for prescribers of buprenorphine for OUD. The MAT Act holds promise to expand access to buprenorphine treatment for people with OUD and helps bring OUD treatment further into the mainstream of health care.

Although empirical evidence supporting the effectiveness of medications for OUD (MOUD), particularly methadone or buprenorphine, as the gold-standard for OUD treatment is widely recognized, MOUD is strikingly underutilized. A recent study (Ronquest et al., 2018) conducted in the US examined patients with OUD who initiated buprenorphine treatment and were followed for 12 months. It was found that, indexed to each patient’s earliest outpatient pharmacy claim for buprenorphine, ‘adherence’ to buprenorphine treatment – defined as the proportion of days covered with buprenorphine being ≥ 0.80 – was associated with reduced odds of return to opioid use and reduced medical costs. However, recent adjusted estimates suggest past-year OUD affected 7631,804 individuals in the US in 2019 (2773 per 100,000 adults 12+), relative to only 1023,959 individuals who received MOUD (365 per 100,000 adults 12+). These data suggest that approximately 86.6% of individuals with OUD nationwide who may benefit from MOUD treatment do not receive it (Krawczyk et al., 2022). Many factors may contribute to suboptimal MOUD use including shortage of MOUD treatment providers and programs, especially in remote and rural areas (Haffajee et al., 2019; Langabeer et al., 2019), waived providers not prescribing at all or to their maximum capacity (Duncan et al., 2020; Krawczyk et al., 2021), stigma harbored and communicated by providers that undermines patients’ engagement in treatment, and lack of integrated systems of care (Jones et al., 2021).

To address the disparity between treatment availability and patients’ access to treatment, on September 20, 2018, the US Surgeon General released Facing Addiction in America: The Surgeon General’s Spotlight on Opioids (US HHS, 2018), which called for a cultural shift in the way Americans talk about the opioid crisis and recommended actions to prevent and treat opioid misuse and promote recovery. On September 28, 2018, the House of Representatives approved comprehensive opioid legislation (H.R. 6, the “Substance Use-Disorder Prevention that Promotes Opioid Recovery and Treatment for Patients and Communities Act,” also known as the “SUPPORT for Patients and Communities Act”). President Trump signed this comprehensive opioid bill into law on October 24, 2018 (SUPPORT, 2018). The law included provisions specific to buprenorphine prescribing, including but not limited to the following: (1) making permanent the prescribing authority for physician assistants and nurse practitioners; (2) permitting clinical nurse specialists, certified registered nurse anesthetists, and certified nurse midwives to become qualifying practitioners for a 5-year period (from October 1, 2018

to October 1, 2023); (3) allowing pharmacies to deliver implantable or injectable medications to treat substance use disorders directly to health care providers and allowing for administration by non-waivered health-care providers; (4) allowing waived practitioners to immediately treat 100 patients at a time if the practitioner is board certified in addiction medicine or addiction psychiatry or if the practitioner provides MOUD in a qualified practice setting, (5) authorizing clinical nurse specialists, certified nurse midwives, and certified registered nurse anesthetists to prescribe MOUD for 5 years, and (6) enabling qualified physicians to prescribe MOUD for up to 275 patients. If a non-physician practitioner (e.g., nurse) works in a state that requires he/she prescribes under the supervision of, or in collaboration with, a qualified physician, the physician does not need a waiver, but needs to be qualified to apply for it.

2. The early stages of a partnership between the US government and private industry

In the 1960s, Reckitt & Colman (R&C) based in Kingston upon Hull in the East Riding of Yorkshire, England, had a product called Codis®, which was a formulation containing aspirin and codeine. That product was the fulcrum for an opioid discovery initiative aimed at identifying an “improvement on codeine” with fewer side effects (e.g., constipation, respiratory depression, abuse liability) than other opioids (Lewis, 1999). The lead candidate molecule that resulted from this development program was buprenorphine (M6029) in 1966, the first of many milestones in its scientific evolution (see Fig. 1, Panel A). Two chemists who played an important role in the buprenorphine development program were John Lewis and Kenneth Bentley. Dr. Lewis had first met Dr. Bentley in 1950 when he was a freshman at Oxford; he was later a doctoral student of the Nobel prize-winning chemist, Sir Robert Robinson. Dr. Bentley, described as the father of a class of semi-synthetic opioids also known as the “Bentley compounds,” was a postdoctoral scientist at Oxford when Dr. Lewis did graduate work there. Dr. Bentley went on to work for MacFarlan Smith in Edinburgh (the main British producer of opium alkaloids at the time). In 1958, MacFarlan Smith entered into a joint venture with R&C to develop over-the-counter analgesics. In 1963, R&C took over the joint project and in 1965 Dr. Lewis joined the company. R&C subsequently provided buprenorphine throughout the 1970s to scientists at the Addiction Research Center (ARC), then located in Lexington, Kentucky (Campbell and Lovell, 2012).

In 1972, Dr. Lewis presented the pharmacological profile of buprenorphine at the annual meeting of the Committee on Problems of Drug Dependence (CPDD). In 1977, R&C registered Temgesic Injection® (0.3 mg/mL buprenorphine) in the United Kingdom (UK) for treating moderate to severe pain; it was packaged in 1-mL terminally autoclaved glass ampoules (emc, 2020). Soon thereafter in 1978, Dr. Donald Jasinski and colleagues at the ARC described the potential of buprenorphine to be used as a pharmacotherapeutic approach for “narcotic addiction” in a paper published in the *Archives of General Psychiatry* (Jasinski et al., 1978). Study participants were healthy, male federal prisoner volunteers with histories of recurrent addiction to narcotics. The studies were conducted over a 1-year period beginning in June 1975. The review of the protocol and informed consent procedures and the opportunity to leave the study without prejudice were in accord with existing Department of Health, Education, and Welfare regulations for the protection of study participants in biomedical and behavioral research.

In 1979, after a nationwide ban on the use of federal prisoners as research participants made it difficult to find participants for clinical studies in Kentucky (Kolberg, 1995), the National Institute on Drug Abuse (NIDA) moved the ARC clinical research program under the direction of Dr. Jasinski to what was then the Francis Scott Key Medical Center, currently the Johns Hopkins University (JHU) Bayview medical campus in Baltimore, Maryland. That move facilitated collaborations with the JHU Behavioral Pharmacology Research Unit headed by Dr. George Bigelow, including a comparative study of buprenorphine and methadone (Bickel et al., 1988a) and a dose-

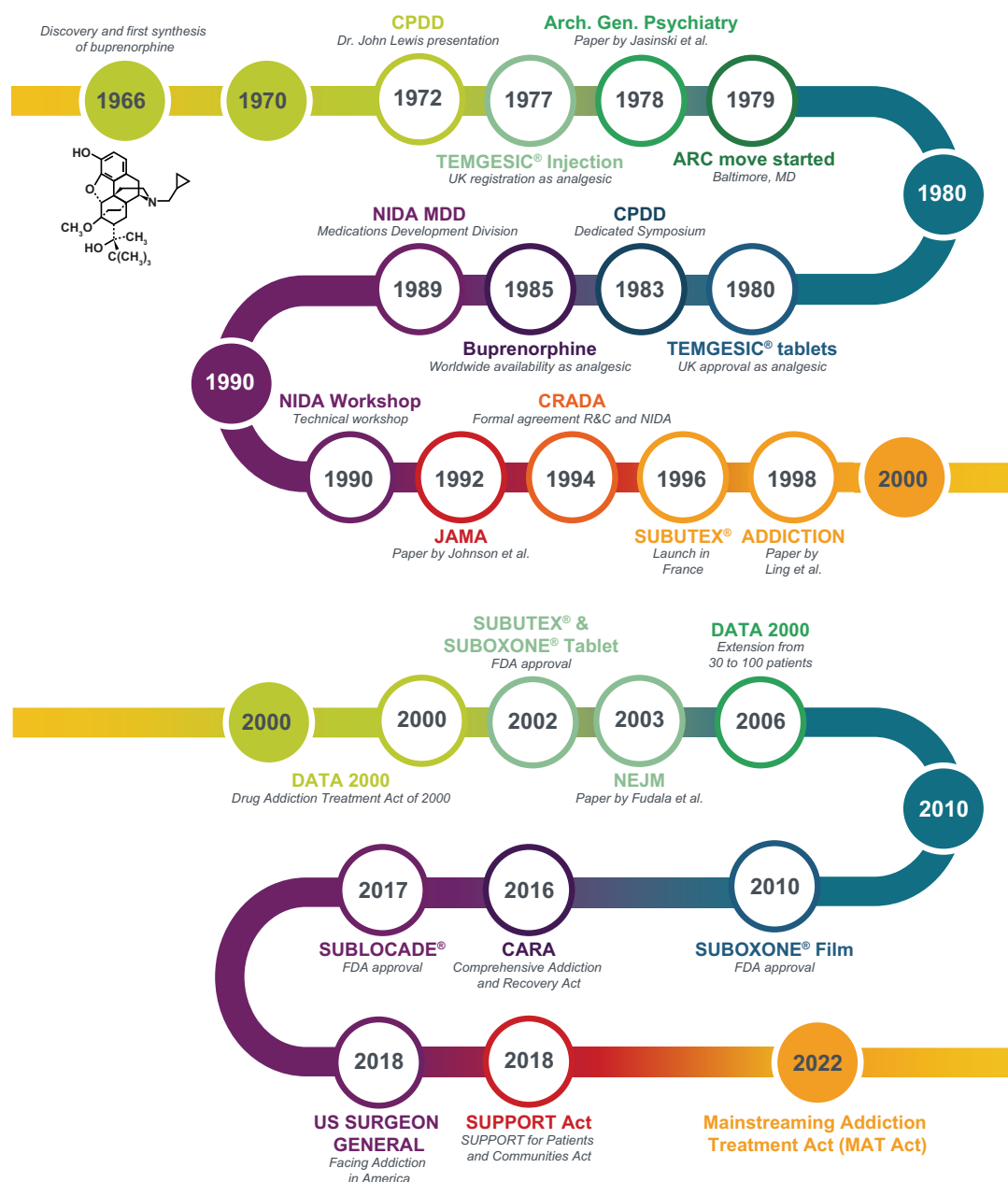


Fig. 1. Key milestones in the development of buprenorphine for the treatment of opioid dependence or opioid use disorder*: Pre-regulatory (Panel A) and post-regulatory (Panel B) approval milestones.

*Opioid use disorder is the current terminology.

ranging study of buprenorphine incorporating an opioid challenge procedure (Bickel et al., 1988b). Dr. Rolley “Ed” Johnson, the pharmacist who had prepared buprenorphine formulations for Dr. Jasinski’s studies in Lexington, assisted Dr. Jasinski in moving the clinical research program from Lexington to Baltimore. While working at the ARC, currently the NIDA Intramural Research Program, Dr. Johnson was lead investigator on numerous studies evaluating the efficacy and safety of buprenorphine. An intense yet informal relationship between R&C and the ARC would facilitate research that antedated a Cooperative Research and Development Agreement (CRADA). R&C continued to supply buprenorphine to NIDA for distribution to its grantees who performed preclinical and clinical studies with buprenorphine. During that period, in February 1983, CPDD held a symposium (Lewis, 1985) that included an important review of buprenorphine presented by Dr. Lewis.

In 1989, the US Congress mandated that a Medications Development Program be established at NIDA. A Medications Development Division (MDD) was created to develop close working relationships between academia, the pharmaceutical industry, and government agencies, including the FDA (Vocci and Ling, 2005). On July 16 and 17, 1990, Dr. Charles R. Schuster, the NIDA Director, convened a technical review meeting on the status of buprenorphine as a potential treatment for opioid dependence. NIDA grantees and researchers presented and discussed the results from clinical pharmacology studies and clinical trials involving buprenorphine used for opioid detoxification or maintenance treatment. Perhaps the most compelling presentation from this body of research was a preliminary report of a study conducted by Johnson and colleagues that was later published in *JAMA* (Johnson et al., 1992). These investigators reported that the group of opioid-dependent, heroin-using participants randomized to 8 mg/day sublingual buprenorphine

solution used significantly less opioids and had better retention than the group randomized to 20 mg of oral methadone, and that buprenorphine was as effective as methadone 60 mg/day. This study ($N = 162$; 70% male; outcomes by sex not reported) was ultimately one of three pivotal clinical trials that the FDA relied on to support the efficacy and safety of buprenorphine for the treatment of opioid dependence.

3. The development of buprenorphine as a drug product

As a result of Dr. Schuster's technical review meeting, a project team was formed in September 1990 to focus on the development of an inter-agency agreement (3Y01-DA-30,011-04) with the Department of Veterans Affairs Cooperative Studies Program (VACSP). The VACSP required that a VA investigator be involved in clinical studies run under the NIDA/VACSP agreement. Dr. Charles O'Brien agreed to fill that role and set up a medications development group at the Philadelphia VA that worked closely with NIDA and the VACSP Data Coordinating Center at Perry Point, Maryland and the VACSP Clinical Research Pharmacy Coordinating Center in Albuquerque, New Mexico. In 1995, Dr. Paul Fudala became the principal investigator of that NIDA/VACSP agreement and continued in that role until 2005. The VACSP collaborated with NIDA on two of the three pivotal studies that helped secure FDA approval of buprenorphine (SUBUTEX®; Reckitt Benckiser Healthcare (UK) Ltd.; Hull, UK) and buprenorphine/naloxone (SUBOXONE®; Reckitt Benckiser Healthcare (UK) Ltd.; Hull, UK) tablets for the treatment of opioid dependence in 2002. The clinical development program was also influenced by work performed by NIDA grantees. For example, Dr. Walter Ling had conducted a large, multisite study of buprenorphine in an opioid-dependent population, and was awarded a supplement to expand the size of the study. NIDA also applied VACSP and other resources to the development and execution of that trial, which explored a range of dosages of buprenorphine as recommended by the technical review meeting participants. The multisite study of opioid-dependent, heroin-using individuals ($N = 736$; 68% male; outcomes by sex not reported) evaluated the safety and efficacy of an 8 mg per day dosage of buprenorphine (given as a sublingual solution) in the maintenance treatment of heroin-dependent individuals compared to a 1 mg per day dosage over a 16-week treatment period (Ling et al., 1998). As a secondary objective, outcomes were determined concurrently for participants treated with dosages of 4 mg and 16 mg per day. Outcomes in the 8 mg group were significantly better than in the 1 mg group on the four domains of retention in treatment, illicit opioid use, opioid craving, and global ratings by participants and staff. In addition to the Johnson et al., (1992) study, these research efforts informed the likely effective dosage range of buprenorphine for opioid-dependence treatment.

Given that the two previous pivotal studies had been performed with a buprenorphine solution, the FDA advised that a single study of the buprenorphine mono and combination tablets as well as a control (placebo) group would be needed to establish the efficacy of the tablet formulations. That study (Fudala et al., 2003), led by principal investigators Drs. Paul Fudala and T. Peter Bridge (representing the VACSP and NIDA, respectively), would be the third study to be considered pivotal with regard to the eventual FDA approval of SUBUTEX® and SUBOXONE® tablets and the only pivotal study that was conducted with tablet formulations rather than a buprenorphine sublingual solution. During the first 4-week part of the study, 326 participants were randomized to daily office-based treatment with buprenorphine (16 mg), buprenorphine/naloxone (16 mg/4 mg) or placebo; 323 (65% male; outcomes by sex not reported) received at least one dose of study medication. Those receiving buprenorphine were inducted with the buprenorphine mono product (8 mg on day 1 and 16 mg on day 2); those randomized to receive buprenorphine/naloxone received buprenorphine on days 1 and 2 and were switched on day 3 to 16 mg/4 mg buprenorphine/naloxone. Participants randomized to the placebo group were given one placebo tablet on day 1 and 2 tablets on day 2 and all dosing days thereafter. Participants completing the 4-week efficacy study were given the opportunity

to continue with the second 48-week phase of open-label buprenorphine/naloxone treatment so that additional safety data could be collected. To ensure collection of a sufficient amount of safety data, a separate 52-week open-label safety study at four sites was conducted to assess buprenorphine/naloxone doses of up to 24 mg/6 mg per day.

The study data and safety monitoring board and the institutional review board recommended terminating the double-blind portion of the study early because both active treatments (buprenorphine 16 mg/naloxone 4 mg and buprenorphine 16 mg) were found to have greater efficacy than placebo. Results showed that participants in the buprenorphine and buprenorphine/naloxone groups demonstrated significantly less opioid use and craving compared to those who received placebo. A total of 461 participants received open-label treatment using flexible dosing up to buprenorphine 24 mg/naloxone 6 mg per day; 268 of those had participated in the double-blind portion of the study. The percentages of urine samples negative for opioids ranged from 35% to 67%. There were no unexpected safety issues during the study.

4. The cooperative research and development agreement (CRADA) between NIDA and R&C

During the conduct of the Ling et al. multicenter study, NIDA and R&C developed the CRADA entitled *Development of Buprenorphine Products for the Treatment of Opioid Dependence*, which was ultimately signed in April 1994 with Dr. Chris Chapleo from R&C as the principal investigator. The goal of the CRADA was, "To cooperatively design and perform the necessary research and cooperatively analyze the data generated to develop and attempt to gain NDA approval of two buprenorphine products for the treatment of opiate dependence." The first product, a mono substance would be administered "on premises" in clinics registered pursuant to NATA requirements (NATA, 1974). The second development product, buprenorphine combined with naloxone, would be attempted under this CRADA as a potential "take-home" medication. The goal was that such a product, if successfully developed, would "minimize the likelihood of diversion for illicit use by opiate-dependent persons." Naloxone was chosen instead of naltrexone after discussions with the FDA. The FDA opined that misuse of a buprenorphine/naloxone product could precipitate withdrawal in a highly physically-dependent user. FDA was concerned that the misuse of a buprenorphine/naltrexone product would produce a prolonged withdrawal that would need to be treated in hospital emergency departments.

Dr. Frank Vocci, a pharmacologist who had joined NIDA in 1989 from the FDA and who later became Director of the MDD, played a critical role in the CRADA development. Duration of the CRADA was 5 years, during which time NIDA and R&C had the following joint responsibilities: (1) develop buprenorphine and buprenorphine/naloxone combination formulations suitable for sublingual use as detoxification and maintenance agents in the treatment of opioid dependence; and (2) submit an NDA to the FDA for suitable opioid dependence treatment products. R&C bought back its US distribution rights but had to set up a US company (R&C Pharmaceuticals, Inc.), as well as develop an infrastructure (secure warehouse services, import permits, etc.) before entering into the CRADA. The CRADA was subsequently amended four times after its initial signing for various reasons, including to complete requirements related to the NDA and to perform a "best practices" study. The CRADA ended on October 28, 2003.

5. Key clinical pharmacology studies within the scope of the CRADA

A bridging bioavailability/PK study of liquid and solid mono products was necessary to establish the bioavailability of the solid dosage form in reference to the liquid. This was required for future clinical trials to link to previous work, thus allowing the FDA to use relevant studies completed with the liquid dosage form to be considered as "pivotal" studies for approval.

The bioavailability of sublingual buprenorphine was determined by comparing plasma levels obtained following a 2-mg sublingual dose to those found after a 1-mg intravenous (IV) dose in 6 research volunteers (Mendelson et al., 1997a). The absolute bioavailability of sublingual buprenorphine was approximately 29%. The relative bioavailability of the solution to the tablet was also assessed. An 8-mg tablet dose was compared to a 7.7-mg sublingual solution dose, and relative bioavailability of the tablet was found to be approximately 50% (Nath et al., 1999; Schuh and Johanson, 1999).

Several clinical pharmacology studies were also conducted to determine the appropriate dose ratio of buprenorphine to naloxone. Two studies were performed in morphine-stabilized (15 mg given intramuscularly [IM] 4 times daily) opioid-dependent participants. Fudala et al., (1998) gave IV injections of placebo, morphine 15 mg, buprenorphine 2 mg, buprenorphine 2 mg/naloxone 0.5 mg, and naloxone 0.5 mg at 48- to 72-hour intervals. Both naloxone and buprenorphine/naloxone produced significant opioid-withdrawal effects compared to placebo. The authors suggested that the addition of naloxone to buprenorphine in a 4:1 ratio would limit the potential for IV abuse in opioid-dependent individuals. Mendelson et al., (1999) also stabilized opioid-dependent participants on 60 mg/day morphine given IM as 4 divided doses. The following challenges were given IV utilizing a double-blind, placebo-controlled, balanced crossover design: buprenorphine 2 mg, buprenorphine 2 mg in combination with naloxone at doses of 0.25 mg (8:1), 0.5 mg (4:1), and 1 mg (2:1), morphine 15 mg, and placebo. The authors reported that the 4:1 and 2:1 ratios of buprenorphine/naloxone produced moderate to high withdrawal and reduced the pleasurable effects of buprenorphine as well as its estimated street value. The effects of the administration of buprenorphine in combination with naloxone to methadone-maintained patients (40–60 mg/day) were also determined (Mendelson et al., 1997b). Naloxone (0.1 mg) produced severe opioid withdrawal. The combination of buprenorphine/naloxone (0.2 mg/0.1 mg) precipitated a withdrawal syndrome at least as severe as the one precipitated by naloxone.

The Fudala et al., (1998) and Mendelson et al., (1999) studies clearly established that 4:1 buprenorphine/naloxone was the optimal dose ratio. Therefore, the decision was made to develop 2-mg and 8-mg buprenorphine tablets and 2-mg/0.5-mg and 8-mg/2-mg buprenorphine/naloxone tablets.

5. The regulatory approval of buprenorphine for the treatment of OUD

Fig. 1. Panel B illustrates key milestones toward regulatory approval of buprenorphine formulations.

5.1. Buprenorphine and buprenorphine/naloxone sublingual tablet

The year after the CRADA was signed, a contaminated blood transfusion scandal and the AIDS epidemic in France incentivized the French regulatory authorities to approve buprenorphine for the treatment of opioid dependence, partly from a harm-reduction perspective with regard to decreasing the risks of HIV and hepatitis infections (Campbell and Lovell, 2012; Feroni and Lovell, 2007). SUBUTEX® first received marketing approval in France in July 1995 and was subsequently launched there in February 1996 by Schering-Plough under license from R&C. Over the next several years, the studies required for the NDA submissions were completed and NDAs for the buprenorphine and buprenorphine/naloxone tablet formulations were finalized and submitted to the FDA. Buprenorphine (SUBUTEX®) and buprenorphine/naloxone (SUBOXONE®) tablets for sublingual administration were approved for the treatment of opioid dependence in the US on October 8, 2002.

5.2. Buprenorphine/naloxone sublingual film

In 2007, Reckitt Benckiser Pharmaceuticals, Inc. (RBP), later to become Indivior Inc., established a partnership with MonoSol Rx (Monosol) (now Aquestive Therapeutics) to develop SUBOXONE® sublingual film by leveraging the MonoSol Rx PharmFilm® technology (Aquestive; Warren; New Jersey). On August 30, 2010, the FDA approved SUBOXONE® sublingual film. Both the SUBUTEX® tablets and SUBOXONE® tablets (no longer marketed in the US) and film are listed on Schedule III of the US Controlled Substances Act (1970) and therefore may be used to treat opioid dependence under DATA 2000 (details can be found in Title 21 US Code (USC) Controlled Substances Act). Since 2010, several transmucosal products have been approved by the FDA for sublingual or buccal use; these include ZUBSOLV® (buprenorphine/naloxone) sublingual tablet from Orexo US Inc., Morristown, New Jersey (FDA approval July 3, 2013) (US FDA, n.d.a), BUNAVAIL® (buprenorphine/naloxone) buccal film from BioDelivery Sciences International Ltd, Raleigh, North Carolina (FDA approval June 6, 2014; now discontinued) (US FDA, n.d.b), and CASSIPA® (buprenorphine/naloxone) sublingual film from Teva Pharmaceuticals USA, Fairfield, New Jersey (FDA approval September 7, 2018; now discontinued) (US FDA, n.d.c). Numerous generic versions of transmucosal buprenorphine/naloxone products have also been approved by the FDA for sublingual or buccal use. As indicated in the FDA Orange Book (US FDA, 2022), none of the buprenorphine products described in this paper were discontinued or withdrawn for safety or effectiveness reasons (discontinuation dates not indicated).

5.3. Long-acting formulations of buprenorphine

Concerns about non-adherence to daily medication treatment (potentially leading to withdrawal, craving, and return to opioid use) and the diversion and nonmedical use of transmucosal buprenorphine products served as an impetus for the development of long-acting formulations of buprenorphine.

5.3.1. PROBUPHINE® (buprenorphine) implant for subdermal administration CIII

The first long-acting formulation of buprenorphine to be approved by the FDA (in 2018) was PROBUPHINE®, a surgical implant (Braeburn Pharmaceuticals, Inc., Plymouth Meeting, Pennsylvania). This implant is a solid polymer matrix (ProNeura™ technology from Titan Pharmaceuticals Inc., New York, New York) consisting of ethylene vinyl acetate and buprenorphine designed to deliver stable buprenorphine concentrations (via continuous dissolution of the matrix) over 6 months. Four cylindrical, matchstick-sized rods (26 mm X 2.5 mm), each containing 74.2 mg of buprenorphine (equivalent to 80 mg of buprenorphine HCl; total buprenorphine dose, 296.8 mg for 4 rods), are implanted subdermally inside the upper arm under local anesthesia. The implants do not biodegrade and must be removed after 6 months. If the patient chooses to continue treatment, implants are then placed in the alternate arm. Healthcare providers must successfully complete a live training program on the insertion and removal procedures and become certified in the PROBUPHINE® Risk Evaluation and Mitigation Strategy (REMS) program prior to inserting or prescribing PROBUPHINE® implants. As a prerequisite for participating in the live training program leading to certification, the healthcare provider must have performed at least one qualifying surgical procedure in the last 3 months. Qualifying procedures are those performed under local anesthesia using aseptic technique and include, at a minimum, making skin incisions, or placing sutures (Titan Pharmaceuticals, Inc., 2022).

In an open-label study conducted in opioid-dependent individuals who had been receiving sublingual buprenorphine prior to it being discontinued, participants were treated with either 2 or 4 PROBUPHINE® rods (dose per rod, 90 mg) (White et al., 2009). This open-label study was followed by a placebo-controlled, randomized trial that evaluated

the efficacy of PROBUPHINE® in combination with individual drug counseling (Ling et al., 2010). Individuals diagnosed with opioid dependence were randomized (2:1) to receive PROBUPHINE® ($n = 108$) or placebo ($n = 55$). Participants could also receive sublingual buprenorphine if needed. The primary endpoint was the percentage of urine samples negative for illicit opioids for weeks 1 to 16 and weeks 17 to 24. Participants who received active buprenorphine implants had a mean percentage of opioid-negative urine samples of 40.4% compared to 28.3% for those who received placebo implants. The percentage of participants who completed the study in the active group was more than twice that of the placebo group (65.7% vs 30.9%). Implant site reactions were the most common adverse events, experienced by 56.5% and 52.7% of the active and placebo groups, respectively. Subsequently, a phase III, 24-week, randomized, double-blind, double-dummy trial found that 85.7% of participants in the active PROBUPHINE® group (placebo sublingual tablets) maintained opioid abstinence relative to 71.9% of those in the sublingual tablet group (placebo PROBUPHINE®) (Rosenthal et al., 2016). Implant-related adverse events occurred in 23% of the active implant group and 13.5% of the active sublingual group, respectively. It is noteworthy that unblinded supplemental sublingual buprenorphine use was allowed as needed during this study and that four buprenorphine implants (totaling 296.8 mg buprenorphine) were expected to yield plasma buprenorphine concentrations (0.5–1.0 ng/mL) at a range comparable with 8-mg/day or less of sublingual buprenorphine. Furthermore, study participants were clinically stable prior to being randomized to a study treatment, having received sublingual buprenorphine for at least 24 weeks at a stable dosage of 8 mg/day or less, and having shown no evidence of opioid withdrawal or illicit opioid-positive urine samples for at least 90 days prior to study entry. Relatedly, the PROBUPHINE® prescribing information indicates that PROBUPHINE® is indicated for maintenance treatment of opioid dependence in patients who have achieved sustained prolonged clinical stability on low-to-moderate doses of a transmucosal buprenorphine-containing product (i.e., doses of no more than 8 mg/day of SUBUTEX® or SUBOXONE® sublingual tablet or generic equivalent). Additionally, patients should not be tapered to a lower dose for the sole purpose of transitioning to PROBUPHINE®. Following a Psychopharmacologic Drugs Advisory Committee Meeting held on January 12, 2016 (Casetext, 2015), PROBUPHINE® received FDA approval on May 26, 2016. On October 16, 2020, Titan Pharmaceuticals Inc. announced its decision to discontinue sales of PROBUPHINE® for OUD in the US as part of its restructuring plan.

5.3.2. SUBLOCADE® (buprenorphine extended-release) injection, for subcutaneous use (CIII)

In 2009, Indivior initiated a new development program (RBP-6000, BUP-XR, or SUBLOCADE®, Indivior Inc., North Chesterfield, Virginia) aimed at identifying safe and effective dosing regimens of a buprenorphine depot product, dissolved in a biocompatible solvent. When the liquid polymer system is injected into the body using a conventional syringe with a needle, it solidifies upon contact with aqueous body fluids to form a solid implant. When a drug (in this case buprenorphine) is incorporated into the polymer solution, it becomes incorporated within the polymer matrix as it solidifies and is slowly released as the polymer biodegrades.

The SUBLOCADE® development program utilized knowledge of the relationship between buprenorphine plasma levels, whole-brain mu-opioid receptor occupancy (MOR), and the key clinical pharmacodynamic (PD) effects of withdrawal suppression and opioid blockade, based on sublingual dosing studies. The primary hypothesis based on preliminary observations in heroin-dependent individuals was that at least 70% brain MOR occupancy by buprenorphine is required to block the subjective drug-liking effects (i.e., to provide opioid blockade) of opioid agonists. Analysis of MOR occupancy (measured with [¹¹C]carfentanil positron emission tomography [PET] imaging) and buprenorphine plasma concentrations suggested that opioid block-

ade would require buprenorphine plasma concentrations ≥ 2 ng/mL (Greenwald et al., 2003). A subsequent study showed that the duration of PD action (estimated by suppression of opioid withdrawal and craving, and blockade of high-dose hydromorphone agonist effects) during 3-day discontinuation following daily sublingual buprenorphine (16 mg/day) dosing decreased over time and was highly correlated with plasma concentrations of buprenorphine and MOR occupancy (Greenwald et al., 2007). Opioid agonist effects produced by hydromorphone were blocked at 4 h after sublingual buprenorphine 16 mg, but returned over time, together with withdrawal symptoms and craving, as plasma concentrations of buprenorphine and MOR occupancy decreased.

The clinical development plan for SUBLOCADE® consisted of a first-in-human, single-dose (20 mg) study, a single ascending-dose study (50, 100 and 200 mg), and a multiple-dose study (50, 100, 200, and 300 mg). Several population PK models were developed using data from opioid-dependent participants who received subcutaneous doses of SUBLOCADE®. In addition, PK/PD models were developed using MOR occupancy data to predict the clinical efficacy of SUBLOCADE® after repeated doses. Results of the population PK analyses jointly with the predicted level of MOR occupancy confirmed earlier findings of a non-linear relation between buprenorphine plasma concentrations and MOR occupancy (Greenwald, 2006), and provided quantitative criteria for the clinical dose selection for SUBLOCADE®; a dose of 300 mg every 28 days was deemed appropriate to immediately achieve effective plasma concentrations of buprenorphine after the first subcutaneous injection and to maintain these effective levels of exposure during chronic treatment (Nasser et al., 2007; Laffont et al., 2016).

An opioid blockade study (Nasser et al., 2016; Indivior Inc., 2022) was then conducted to test the original “blockade of subjective opioid effects” hypothesis. Stabilization doses of sublingual buprenorphine (considered collectively and ranging from 8 to 24 mg across participants) prior to injection of SUBLOCADE® failed to fully block drug liking of hydromorphone 18 mg IM. In contrast, complete hydromorphone blockade of drug-liking effects was achieved by SUBLOCADE® for both hydromorphone doses (6 and 18 mg) during each week of testing for the 4 weeks after the first dose of SUBLOCADE® and also throughout the 8 weeks following the second SUBLOCADE® injection. In a [¹¹C]carfentanil PET study of SUBLOCADE® in two participants with OUD (one receiving 200-mg injections and one receiving 300-mg injections), 75% to 92% occupancy of whole-brain MOR was maintained for 28 days following the last dose under steady-state conditions. These data strongly supported the hypothesis that the minimum threshold plasma concentration of buprenorphine needed to effectively block the subjective drug-liking effects of a full opioid agonist such as hydromorphone was 2 ng/mL, which translated into at least 70% MOR occupancy in the brain for the entire 1-month period. Analysis of a secondary outcome (drug vs. money choices) also found that hydromorphone-seeking behavior was significantly attenuated throughout the monthly interval. This measure is not often included in FDA qualifying trials; the finding was, therefore, encouraging for efficacy and reinforces the predictive validity of liking as an index of abuse potential.

The efficacy and safety of SUBLOCADE® was subsequently assessed in a phase III, randomized, double-blind, placebo-controlled, multicenter trial with two active dose regimens versus placebo ($n = 504$) as well as in a phase III, open-label, long-term safety and tolerability trial with flexible dosing ($n = 669$). SUBLOCADE® was shown to produce significantly higher percentage abstinence and treatment success versus placebo; treatment outcomes were consistent across other efficacy endpoints including control of opioid craving and withdrawal symptoms (Haight et al., 2019). SUBLOCADE® in combination with individual counseling was shown to be safe and well-tolerated and had higher medication satisfaction scores compared with participants who received placebo and individual counseling. Importantly, no compensatory non-opioid illicit drug use was observed with SUBLOCADE® treatment. Exposure–response relationships among clinical endpoints

and predicted whole-brain MOR occupancy confirmed that the overall probability of no illicit opioid use (i.e., abstinence) increased within the range of 70% to 90% MOR occupancy. The probability of zero craving also increased between 70% to 90% MOR occupancy. The results indicated that 4 monthly SUBLOCADE® 100-mg doses following 2 initial doses of 300 mg were associated with a mean average buprenorphine plasma concentration (C_{avg}) of 3.14 ng/mL, which corresponded to 75% MOR occupancy. Additionally, the results indicated that 6 monthly doses of SUBLOCADE® 300 mg provided a buprenorphine mean plasma C_{avg} of 6.32 ng/mL, which corresponded to 83% MOR occupancy. A recent analysis of phase II and phase III SUBLOCADE® data employed an integrated PK/PK model to characterize buprenorphine exposure-response relationships. Buprenorphine plasma concentrations were modeled in relation to several key clinical endpoints (suppression of withdrawal and craving, and blockade of opioid liking and drug-seeking behavior). This analysis confirmed that concentrations in the 2–3 ng/mL range are effective in producing broad clinical benefits on these measures (Laffont et al., 2022). This comprehensive analysis further identified several individual differences in these PK/PD relationships, including demographic, genetic, and social factors, which are an important topic of future study for optimizing outcomes.

The long-term safety of SUBLOCADE® was assessed in a phase III, open-label, multicenter study (Andorn et al., 2020) in adults with moderate or severe OUD. This study enrolled 257 participants from the previously-conducted placebo-controlled, double-blind phase III trial (rollover group) and 412 new participants who had not previously been treated with SUBLOCADE® (*de novo* group). Overall, 66.8% of participants reported >1 treatment-emergent adverse event (TEAE). Injection-site TEAEs (13.2% of participants) were mostly mild or moderate in severity. There were no clinically meaningful changes in safety assessments. The incidence of TEAEs, including injection-site TEAEs, were lower during the second 6 months of treatment versus the first 6 months (descriptive comparison of double-blind and open-label populations). The percentage of participants abstinent after 12 months of treatment was 61% in the rollover group and 75.8% in the *de novo* group. An additional 6-month, open-label, flexible-dose study (12 months + 6 months for a total of 18 months) included 208 individuals who had completed the open-label phase III trial. Participants received an initial injection of SUBLOCADE® 300 mg and subsequent monthly 300-mg or 100-mg flexible doses. In the 6-month extension study, 90.4% were abstinent at the 18-month timepoint.

SUBLOCADE® was granted US Fast Track Designation on May 23, 2016. A pre-NDA meeting was held on December 14, 2016, which was followed by the NDA submission on May 30, 2017, and Prescription Drug User Fee Act (PDUFA) Priority Review designation on July 29, 2017. A joint advisory committee of the Psychopharmacologic Drugs Advisory Committee and the Drug Safety and Risk Management Advisory Committee was held on October 31, 2017, to discuss the efficacy, safety, and benefit-risk profile of SUBLOCADE®. SUBLOCADE® (buprenorphine extended-release injection for subcutaneous use [CIII]) was approved by the FDA on November 30, 2017, for the treatment of moderate to severe OUD in patients who have initiated treatment with a transmucosal buprenorphine-containing product followed by dose adjustment for a minimum of 7 days.

Beyond clinical efficacy and safety, patient-reported outcomes (PRO) research was performed in accordance with the 2009 FDA Guidance for Industry, *Patient-Reported Outcome Measures: Use in Medical Product Development to Support Labeling Claims* (US FDA, 2009) and the 2018 FDA Draft Guidance for Industry, *Opioid Use Disorder: Endpoints for Demonstrating Effectiveness of Drugs for Medication-Assisted Treatment* (US FDA, 2020). In a 24-week, phase III, placebo-controlled study, participants receiving up to 6 monthly injections of SUBLOCADE® 300/300 mg (6 × 300 mg) and 300/100 mg (2 × 300 mg then 4 × 100 mg) reported better health, increased medication satisfaction, increased employment, and decreased healthcare utilization compared with participants receiving placebo injections (Ling et al., 2019a). In

an open-label study, participants who received SUBLOCADE® for up to 12 months experienced improvement in health-related quality of life, health status, treatment effectiveness as assessed by the four domains (substance use, health, lifestyle, community) of the Treatment Effectiveness Assessment tool and high medication satisfaction (Ling et al., 2019b). These improvements were sustained over the study period, suggesting that SUBLOCADE® facilitates long-term engagement of life activities that reflect recovery (Ling et al., 2020a). Finally, the RECOVER (Remission from Chronic Opioid Use—Studying Environmental and socioeconomic Factors on Recovery) study (Ling et al., 2019c) is an ongoing multisite, non-interventional cohort study examining the long-term recovery process among individuals with moderate to severe OUD who received one or more doses of SUBLOCADE® in the phase III clinical trials. The objectives of the RECOVER study are to characterize periods of abstinence over a 24-month observational window, examine the clinical, environmental, and socioeconomic factors associated with periods of abstinence and return to opioid use, and understand the persistent effects of MOUD on clinical, humanistic, and economic outcomes in persons with OUD. The RECOVER data reported so far indicated that 75.3% of participants who had been treated with SUBLOCADE® for 12 months achieved sustained illicit opioid-free weeks for the entire first 12 months versus only 24.1% of participants who were treated with SUBLOCADE® for 0–2 months. Improved and sustained employment status and health-related quality of life over the 12-month observational period were also reported (Ling et al., 2020b). A recent study confirmed that up to 18 months after the last SUBLOCADE® injection, RECOVER study participants largely retained or improved their outcomes over time, almost half of them reported sustained abstinence for the 18-month period, and those receiving a longer duration of SUBLOCADE® had better outcomes (Boyett et al., 2022). A follow up study (Craft et al., 2022) at an average of 4.2 years post-treatment showed high levels of abstinence from opioids (74.1% in the past 7 days and 60% in the past 30 days). Furthermore, lower odds of opioid misuse were associated with improved quality of life and treatment effectiveness whereas higher odds of opioid misuse were associated with depression, psychological distress, opioid craving and withdrawal, and OUD symptoms (DSM 5 criteria) (American Psychiatric Association, 2013).

A recent 12-month prospective single-arm, multicenter, open-label trial in Australia, evaluated treatment retention, drug use, and other key outcomes following treatment with SUBLOCADE® among people who were opioid-dependent with previous treatment experience (Farrell et al., 2022). Retention following SUBLOCADE® treatment at 24 and 48 weeks was 86% and 75%, respectively. The odds of use of all illicit substances except for cannabis use decreased significantly with time retained in treatment. For example, the odds of heroin use in the past month decreased by 20% on average for every four weeks retained in treatment. Similarly, the odds of non-prescribed opioid use reduced by 17%, and the odds of past-month injecting drug use decreased by 26% for every 4 weeks retained in treatment. Concordance of self-reported substance use and urine test results were 94%, 92%, 93%, 88% and 100% for cannabis, amphetamines, opioids, benzodiazepines and cocaine respectively. The odds of being employed increased 58% for every 4 weeks people were retained in treatment. Significant increases in quality of life and treatment satisfaction were also observed together with a moderate decline in pain and a significant reduction in the odds of moderate-severe depression for every 4 weeks retained in treatment.

A pilot proof-of-concept study (Lee et al., 2021) also compared treatment retention following release from jail among adults receiving daily sublingual buprenorphine-naloxone vs. those receiving SUBLOCADE® and showed that participants in the SUBLOCADE® arm had fewer jail medical visits compared with daily sublingual medication administration. Community buprenorphine treatment retention at week 8 post-release was 69.2% in the SUBLOCADE® group vs. 34.6% in the sublingual buprenorphine-naloxone group, and rates of opioid-negative urine tests were 55.3% in the SUBLOCADE® group vs. 38.4% in the sublingual buprenorphine-naloxone group. There were no differences in

rates of serious adverse events, no overdoses, and no deaths. Although preliminary, these findings suggest that monthly injectable extended-release buprenorphine is a useful OUD treatment option before and after release from jail. This conclusion is also supported by a recent series of in-depth qualitative interviews conducted among adult participants with OUD who were recently released from New York City jails and maintained on SUBLOCADE® after switching from daily sublingual buprenorphine-naloxone (Cheng et al., 2022). SUBLOCADE® treatment during jail-to-community reentry was perceived by participants as a favorable treatment approach to reduce their risk of opioid reuse, illicit activities, stigma associated with daily receipt of MOUD, and COVID-19 exposure. In another pilot study across two Veterans Health Administration outpatient clinics, 26 clinically complex patients at high risk for hospitalization and mortality received at least one SUBLOCADE injection between December 1, 2018 and April 1, 2020 (Cotton et al., 2022). Treatment retention was robust, 81% received more than 6 injections and 77% were maintained on the 300 mg dosage strength. Treatment was also associated with reduction in Emergency Department (ED) visits (-48%), hospitalizations (-39%), days of hospitalization (-41%), non-prescribed opioid use, and homelessness.

A recent clinical pharmacology study also assessed the competitive interaction of fentanyl and buprenorphine on respiratory depression (Moss et al., 2022). The study evaluated the ability of treatment-relevant plasma concentrations of buprenorphine to prevent respiratory depression as measured by minute ventilation (VE) and resultant apnea (cessation of breathing) induced by escalating doses of intravenous fentanyl. The maximum decrease in VE, induced by the highest dose of fentanyl, was nearly 60 percent less with a steady-state plasma buprenorphine concentration of 2.0 ng/mL compared with placebo (33.7 percent vs 82.3 percent decrease in VE, respectively). The risk of experiencing apnea requiring verbal stimulation after fentanyl dosing was significantly lower with buprenorphine than with placebo. A follow-up publication further characterized buprenorphine-fentanyl interaction at the level of the MOR (Olofsen et al., 2022). The effects of escalating intravenous fentanyl doses within the 0.25–0.70 mg/70 kg range in chronic opioid users on isohypercapnic ventilation at 2–3 background doses of buprenorphine (target plasma concentrations range: 0.2–5 ng/mL) were quantified using receptor association/dissociation models combined with biophase distribution models. The results showed that high doses of fentanyl caused pronounced respiratory depression and apnea. In contrast, when combined with fentanyl, buprenorphine produced a receptor binding-dependent reduction of fentanyl-induced respiratory depression. In chronic opioid users, a protective effect against high-dose fentanyl was observed at buprenorphine plasma concentrations ≥ 2 ng/mL, suggesting that when buprenorphine MOR occupancy is sufficiently high, fentanyl is unable to activate the MOR and consequently will not cause further respiratory depression in addition to the mild respiratory effects of buprenorphine.

5.3.3. *BRIXADI™ (buprenorphine) extended-release injection for subcutaneous use*

Another sustained-release subcutaneous buprenorphine formulation (weekly and monthly depots), CAM2038 (BRIXADI™), was developed using FluidCrystal® Injection Depot Technology (Camurus AB, Lund, Sweden). Camurus and Braeburn Pharmaceuticals Inc. submitted the NDA for BRIXADI™ weekly and monthly depots on July 19, 2017. PDUFA Priority Review designation was granted on September 18, 2017. A joint advisory committee of the Psychopharmacologic Drugs Advisory Committee and the Drug Safety and Risk Management Advisory Committee was held on November 1, 2017 to discuss the efficacy, safety, and benefit-risk profile of BRIXADI™ weekly and monthly depots (Braeburn Pharmaceuticals, Inc., 2018). On December 21, 2018, Braeburn Pharmaceuticals, Inc., received FDA's tentative approval of BRIXADI™. However, Braeburn Pharmaceuticals, Inc., received a first complete response letter from the FDA in December 2020 and a second complete response letter in December 2021; BRIXADI™ is not currently

approved in the US. On December 8, 2022, Braeburn Pharmaceuticals, Inc. announced that resubmission of the NDA for BRIXADI™ was accepted by the FDA with a PDUFA action date set for May 23, 2023 (Braeburn Inc., 2022). BRIXADI™ is the US trademark for Camurus' product BUVIDAL® (buprenorphine prolonged-release solution for subcutaneous injection in prefilled syringe) which was approved for the treatment of OUD in Europe and Australia.

The FluidCrystal® Injection Depot Technology (Camurus AB, Lund, Sweden) comprises a lipid-based liquid with a dissolved active ingredient. Once injected, the lipid solution transforms into a liquid crystalline gel that encapsulates buprenorphine and releases it as the depot biodegrades. BRIXADI™ was developed in different fixed-dose compositions. The clinical development plan for BRIXADI™ consisted of four PK studies of weekly and monthly BRIXADI in healthy volunteers or patients with OUD (Albayaty et al., 2017; Haasen et al., 2017), a phase II opioid-blockade study (Walsh et al., 2017), a 24-week, phase III, randomized, double-blind, double-dummy study of BRIXADI™ versus daily sublingual buprenorphine/naloxone (Lofwall et al., 2018) and a 48-week, phase III, open-label, long-term safety study of BRIXADI™ in new-to-treatment patients and patients switched from daily sublingual buprenorphine/naloxone (Frost et al., 2019).

The phase II opioid-blockade study (Walsh et al., 2017) examined the efficacy of once-weekly BRIXADI™ at two doses, 24 mg and 32 mg, to block the subjective drug-liking response to acute IM hydromorphone (6 and 18 mg) and to suppress opioid withdrawal in a non-treatment-seeking OUD population. Both doses of weekly BRIXADI™ suppressed responses to hydromorphone and opioid withdrawal over a 14-day period (two weekly injections).

The phase III, double-blind, double-dummy study (Lofwall et al., 2018) randomized 428 adults with moderate-to-severe OUD to flexible dosing with weekly and monthly BRIXADI™ or daily sublingual buprenorphine/naloxone. Primary endpoints were non-inferiority in the proportion of opioid-negative urine samples and responder rate. A responder was defined as having no evidence of illicit opioid use at nine pre-specified time points. Supplemental weekly 8 mg subcutaneous injections of buprenorphine were allowed in both treatment groups. Non-inferiority between BRIXADI™ and sublingual buprenorphine/naloxone was demonstrated for both primary endpoints. The safety profile of BRIXADI™ was generally consistent with the known safety profile of buprenorphine with the exception of mild-to-moderate injection-site adverse events.

The long-term safety of BRIXADI™ weekly (8, 16, 24, or 32 mg) and monthly (64, 96, 128, or 160 mg) depots was assessed in a phase III, open-label, observational, multicenter, 48-week trial (Frost et al., 2019) in 227 adults (143 males and 84 females) with OUD who received BRIXADI™. Participants in this study were allowed to enter in two ways: 37 participants (16.3%) initiated treatment with BRIXADI™ and 190 participants (83.7%) converted from sublingual buprenorphine. All participants receiving BRIXADI™ were allowed dose adjustments as needed weekly at scheduled visits or with 8 mg BRIXADI™ supplemental injections at unscheduled visits up to a maximum weekly dose of 40 mg per week. Participants receiving BRIXADI™ monthly who needed additional temporary buprenorphine were allowed a maximum of two supplemental injections of 8 mg BRIXADI™ per week; dose adjustments could be made at investigator discretion at scheduled visits.

6. Conclusions

MOUD combined with psychosocial interventions is the most effective treatment option for patients suffering from OUD (ASAM, 2020; Connock et al., 2007; Dugosh et al., 2016). MOUD has been shown to be associated with positive outcomes for reducing drug use, criminal activity (Gisev et al., 2019; Gossop et al., 2000; Oliver et al., 2010; Roislien et al., 2014), risk behaviors and HIV-transmission (Marsch 1998), overdoses, and overall mortality (Gisev et al., 2015;

Sordo et al., 2017) as well as improving rates of retention in treatment (Lim et al., 2022).

As a mu-opioid partial agonist, buprenorphine has a favorable pharmacological safety profile relative to full opioid receptor agonists (Walsh et al., 1994; Soyka 2017), which may improve the therapeutic dosing margin for certain clinical subgroups including pregnant individuals (Brogly et al., 2014; Lee et al., 2019). Rapid induction helps patients to stabilize quickly and maximizes retention in treatment (Spreen et al., 2022; Wong et al., 2021). Although there are challenges to buprenorphine induction in the era of fentanyl, our understanding of the neuropharmacological mechanisms underlying this transition is advancing and can improve the benefit/risk ratio through appropriately selected lower- or higher-dose induction strategies (Greenwald et al., 2022).

The increased availability of multiple FDA-approved formulations of buprenorphine over the past two decades, as described here, offers a promising treatment landscape that affords an unprecedented range of choices for healthcare providers and persons with OUD. Indeed, the proportion of persons with OUD receiving buprenorphine treatment monotonically increased during 2011–2020 from 2.7% to 16.2% (SAMHSA, 2021b). It is hoped that these options will increase the likelihood that patients will enter and stay in treatment to accelerate their recovery. It is only through integrated clinical treatment programs that actively engage patients and their communities that we, as a society, will progressively break down barriers and misconceptions about substance use disorders and their treatment. Nonetheless, systemic barriers that have created “maximally disruptive care” for persons with OUD continue to pose challenges in reducing disparities in access to, and engagement in, treatment (Amiri et al., 2021; Englander et al., 2022); persistent efforts are needed to address this problem. Focusing on the importance of continuity of care across settings and underserved vulnerable populations with OUD, including adolescents (Camenga et al., 2019; Hadland et al., 2017), pregnant individuals (Link et al., 2020; Klamans et al., 2017), those involved in the criminal justice system (Evans et al., 2019), and economically disadvantaged persons, using lower-threshold approaches such as interim treatment (Samsó et al., 2022) and telehealth (Guillen et al., 2022) will become increasingly critical. Furthermore, key aspects of OUD treatment must include monitoring patient progress in the short-, medium-, and long-term, focusing on retention in treatment (Connock et al., 2007; Kennedy et al., 2022; Klimas et al., 2021; Shulman et al., 2021), and providing evidence-based training to healthcare providers (Gugala et al., 2022). Incorporating harm reduction practices into MOUD (Greenwald, 2018; Taylor et al., 2021), integrating care for persons with OUD and co-occurring mental health (Ghabrash et al., 2020) and medical problems (Fanucchi et al., 2019), and understanding better the underlying causes of return to opioid use are equally critical. The magnitude and dynamic nature of the opioid epidemic still requires significant investments to demonstrate that comprehensive drug treatment strategies lead to better outcomes that ultimately offset medical costs associated with substance use disorders and the associated costs of incarceration, shelter, and welfare when these burdensome conditions are untreated.

Contributions

CH contributed to the concept and design of this review, drafting of the manuscript and revising it critically for important intellectual content. PJF and MKG contributed to the drafting of the manuscript and revising it critically for important intellectual content. All authors approved the final version of the submitted manuscript.

Disclosures

CH and PJF are full-time employees of Indivior Inc. MKG has received consulting and speaker fees from Indivior Inc. unrelated to the preparation of this manuscript.

Declaration of Competing Interests

CH and PJF are full-time employees of Indivior Inc. MKG has received consulting and speaker fees from Indivior Inc. unrelated to the preparation of this manuscript.

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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